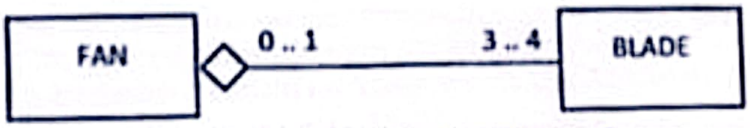
Next 5 questions refer to the following statement:
A document is made of many pages. A page belongs to only one book. A page m characters, tables, pictures. A table is constructed of many cells. A cell is only Cover is a special type of page.

1. The relationship between page and cover is:
   (A) Aggregation
   (B) Dependency
   (C) Composition
   (D) Generalization
2. The relationship between character and cell is:
   (A) Aggregation
   (B) Generalization
   (C) Composition
   (D) None
3. The relationship between character and page is:
   (A) Composition
   (B) Aggregation
   (C) Generalization
   (D) None
4. The relationship between table and cell is:
   (A) Aggregation
   (B) Dependency
   (C) Composition
   (D) Generalization
5. The relationship between book and page is:
   (A) Composition
   (B) Aggregation
   (C) Generalization
   (D) None
6. Which of the following is not a part of the Class Diagram?
   (A) Attributes
   (B) Operations
   (C) Object
   (D) Stereotype
7. What does reflexive association mean?
   (A) Association among objects of two different classes
   (B) Association among class
   (C) Association among objects of two different classes using index
   (D) Association among the same parent
8. Which of the followings addresses the verbs of the domain?
   (A) Structural Things
   (B) Behavioral Things
   (C) Annotational Things
   (D) Grouping Things
9. Which of the following classes is created due to relationship?
   (A) Component
   (B) Active Class
   (C) Interface Class
   (D) Association Class

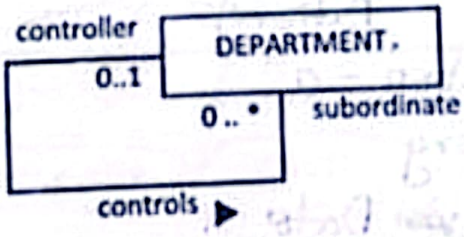
Next 2 questions refer to the following diagram:



29. Which of the following statements is correct based on the above diagram?
    (A) A fan contains maximum 4 blades and minimum 0 blade.
    (B) A fan may have minimum 3 blades; maximum 4 blades.
    (C) A fan contains minimum 4 blades or more
    (D) A fan contains exactly 4 blades.
30. Which of the following statements is correct based on the above diagram?
    (A) A blade is not a part of any fan
    (B) A blade must belong to exactly one fan
    (C) A blade may be a part of maximum one fan or no fan at all
    (D) A blade belongs to minimum 1 fan

Part – B **True/False** **[1x 10 = 10 Marks]**

Use OMR Sheet to answer your questions. Answer all questions.
Darken circle 'A' for TRUE or circle 'B' for FALSE in the OMR Sheet



Question No. 31 – 33 are based on the above diagram.

31. Every department must be controlled by exactly one other department.
32. There may be no subordinate under one department.
33. Every department must be controlled by another department.
34. The maximum multiplicity of the association at the whole side can be more than one in case of composition.
35. Generalization is a top-down process.
36. Objects are templates of classes.
37. Relationship is prerequisite of interaction.
38. Dependency relationships in use case diagram are of two types – include and exclude.
39. Behavioral things address mostly the dynamic part of the system.
40. Association Class stores information about relationship.

10. Which one of the following is not part of a static diagram?
    (A) Association (B) Interaction
    (C) Generalization (D) Aggregation
11. In which of the following phases of RUP the **least** amount of **'Testing'** is carried out?
    (A) Inception (B) Elaboration
    (C) Construction (D) Transition
12. Alpha testing is expected at the end of following phases of RUP:
    (A) Inception (B) Elaboration
    (C) Construction (D) Transition
13. In UML, which of the following type of visibility allows access only inside the class?
    (A) Public (B) Protected
    (C) Private (D) Packaged
14. Super Class – Sub Class relationship known as:
    (A) Generalization (B) Realization
    (C) Aggregation (D) Composition
15. What is the purpose of a use case diagram?
    (A) To represent a structural design view (B) To express how a system performs its tasks for actors
    (C) To address the dynamic interaction view of the system (D) To establish a common understanding with end users
16. The number of objects involved in association relationship is identifies by:
    (A) Association Role (B) Multiplicity
    (C) Association constraint (D) Qualified Association
17. Which of the following represents exchange of messages among objects?
    (A) State Machine (B) Activity
    (C) Interaction (D) Component
18. Which of the following represents whole-part relationship among objects?
    (A) Association (B) Aggregation
    (C) Generalization (D) Dependency
19. Value of an attribute that remains same for all the objects of a class is known as:
    (A) Default value attribute (B) Derived attribute
    (C) Slash (D) Static attribute
20. Which of the following contains **specialized** elements?
    (A) Parent Class (B) Super Class
    (C) Child Class (D) Interface

Match the notation of <u>next 5 questions</u> with the items in the following list:

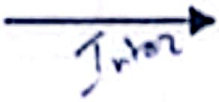
A. Interface B. Note C. Interaction D. Active Class E. Node

21.

22.

23.

24.

25. 

26. Which of the following relationships is not found in a use case diagram?
    (A) Association (B) Dependency
    (C) Generalization (D) Realization
27. Class Diagram represents:
    (A) Static Use Case View (B) Static Design View
    (C) Dynamic Process View (D) Dynamic Interaction View
28. Interaction is a pre-requisite of:
    (A) Behavior (B) Things
    (C) Relationship (D) None